

Problem 4.19

A fund is earning 5% simple interest. If $i_n = 0.04$, calculate n .

Solution

For simple interest,

$$a(t) = 1 + it.$$

Since the fund earns 5% simple interest,

$$i = 0.05.$$

The effective rate in period n is

$$i_n = \frac{a(n) - a(n-1)}{a(n-1)}.$$

Using $a(t) = 1 + it$,

$$i_n = \frac{(1 + 0.05n) - (1 + 0.05(n-1))}{1 + 0.05(n-1)}.$$

Simplifying,

$$i_n = \frac{1 + 0.05n - 1 - 0.05n + 0.05}{1 + 0.05n - 0.05} = \frac{0.05}{0.95 + 0.05n}.$$

Since $i_n = 0.04$,

$$0.04 = \frac{0.05}{0.95 + 0.05n}.$$

Therefore,

$$0.04(0.95 + 0.05n) = 0.05.$$

So,

$$0.95 + 0.05n = \frac{0.05}{0.04} = 1.25.$$

Hence,

$$0.05n = 1.25 - 0.95 = 0.30,$$

and thus

$$n = \frac{0.30}{0.05} = 6.$$

Therefore,

$$\boxed{n = 6}.$$